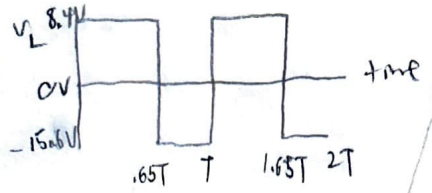
HW 5

Dylan Thomas
ECEN 164

1. $L = 25\mu H$ DC steady $D = .65$ $f_s = 100kHz$
$C = 15mF$ $V_{in} = 24V$ $R = 10\Omega$ $T = 10ms$

a.
$V_o = V_{in}(D)$
$V_o = 24(.65) = 15.6V$
$24 - 15.6 = 8.4V$



avg = 0V

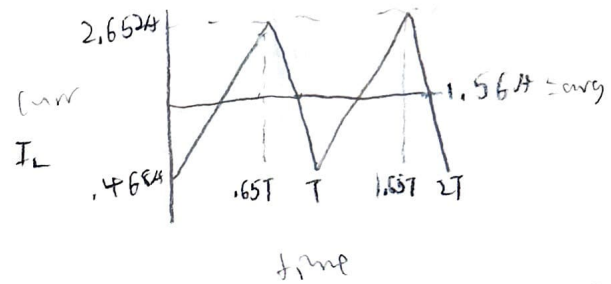
b. $I_o = \frac{15.6}{10} = 1.56A$

$I_{LON} = \frac{(24-15.6).65T}{2\,25\mu H} = 2.184$

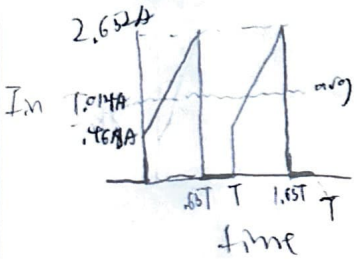
$I_{LOFF} = \frac{-15.6(1-.65)T}{25\mu H} = -2.184$

$1.56 + \frac{2.184}{2} = 2.652A$

$1.56 - \frac{2.184}{2} = .468A$

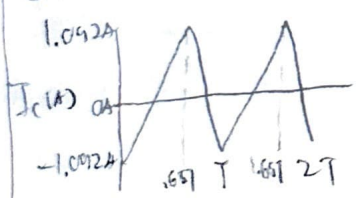


c. (starts at .468)



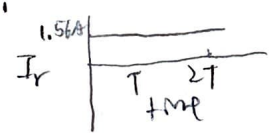
$24(I_{in\,avg}) = 1.56(15.6)$
$I_{in\,avg} = 1.014$
True?
$.468(.65) + \frac{(2.652 - .468)*.65}{2}$
$= 1.014$ ✓ yes
$I_{in\,avg} = 1.014A$

d.



$I_{c\,avg} = 0A$

e.



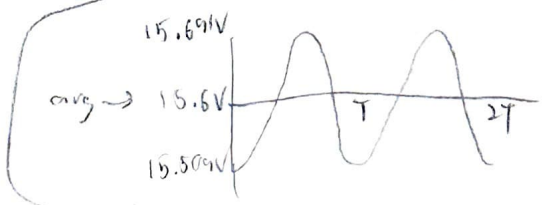
avg = peak = 1.56A

f. $\Delta V_o = \frac{\Delta i_L}{8Cf_s} = \frac{2.184}{8(15m)100k}$

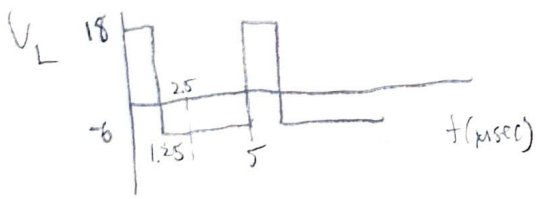
$\Delta V_o = .182$

$\frac{.182}{2} = .091V$

$15.6 + .091 = 15.691$
$15.6 - .091 = 15.509$



2. Buck converter waveform



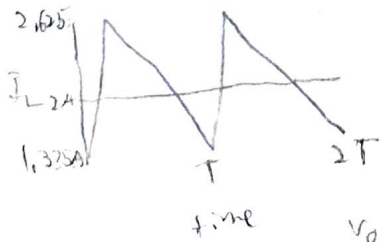
$18(1.25) - 6(3.75)$
$= 0$
a. The inductor is in steady state

b. $L = 18mH$ $V_{in} = 24V$ $V_o = 6V$
$P_{out} = 12W$
$D = .25$
$T = 5ms$

$12W = \frac{6^2}{R}$
$R = 3\Omega$
$\frac{6}{3} = 2A = I_o$

$I_{LOFF} = \frac{-6(1-.25)5ms}{18mH}$
$= 1.25A$

avg = 2A
$\frac{1.25}{2} = .625$



c. $R_{crit} = \frac{6}{\frac{1.25}{2}} = 9.6\Omega$

i. $2 < 9.6$; CCM; $V_o = 6V$

ii. $20 > 9.6$; DCM; would be $V_o = 6V$

$V_o = V_s\left[\frac{2D}{D + \sqrt{D^2 + 2L/RT}}\right]$

$V_o = 24\left(\frac{.5}{.25 + \sqrt{.25^2 + 8(18\times10^{-6})/20(5\times10^{-6})}}\right) = 8.13V$

d. $\Delta V_o = \frac{\Delta i_L}{8 C f_s}$ $\quad \frac{\Delta V_o}{V_o} = .005 \rightarrow \frac{\frac{\Delta i_L}{8 C f_s}}{V_o} = .005 \rightarrow \frac{\frac{1.25}{8C(200k)}}{6} = .005$

$\boxed{C = 26.042 \mu F}$

3.

a. Voltage ripple: Larger f_s = less ripple (good)

Passives: Larger f_s allows for smaller caps and inductors (good)

Energy and losses: Higher f_s means more switching losses and greater power usage (bad)

b. MOS 1 — faster switching

MOS 2 — Lower $R_{DS\,on}$

If care about timing above all else and don't mind power losses, M1 is better. If we care about power loss most, then M2 is better

c. Diode 2 achieves the same current numbers as Diode 1 but w/ about .5 less V_f. Much better for losses. For a buck converter, I don't see a reason to choose D1. D2 clears.

4.

$V_{in} = 10V$

$f_s = 200k$

$I_{in} = 2A$

$V_o = V_{in}(D)$

$5 = 10(D)$

$\boxed{D = \frac{1}{2} = .5}$

$P_{out} = P_{in}$ b/c ideal

$\boxed{P_{out} = 10(2) = 20W}$

$\frac{20}{5} = I_o = 4A$

non ideal MOS $\quad R_{DS\,on} = 50m\Omega$

$V_f = .65V$ ← diode

MOS loss:

$I_o^2 R_{DSON} * D$

$= 4^2 (.05)(.5)$

$= .4W$

Diode loss:

$(1-D) I_o * V_D$

$= (.5)(4)(.65)$

$= 1.3W$

total $= .4 + 1.3 = 1.7W$

Would a synchronous system be better? Yes b/c in that system we have 2 MOSFETS. 2 MOSFETS means less conduction loss.

$2 * .4 < 1.7W$

$.8W < 1.7W$